

The largest airtanker can carry up to **24,000 gallons (91,000 liters)** of retardant, enough to fill **300 bathtubs**.

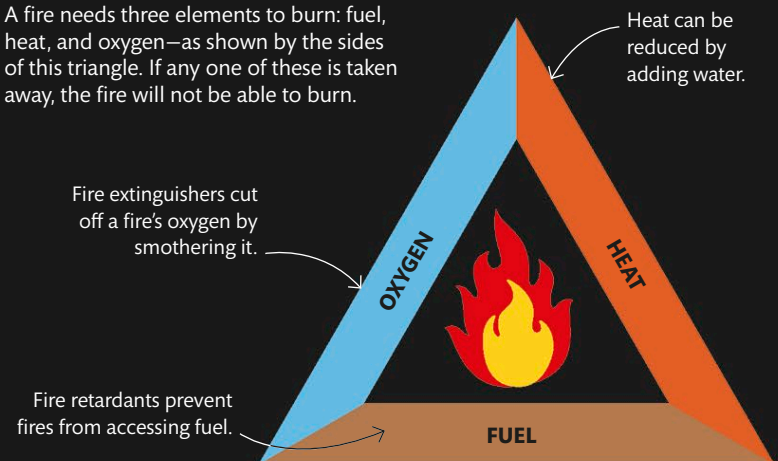
FIGHTING FOREST FIRES

FIRE RETARDANTS

Although forest fires can be a natural part of a forest's life, most forest fires today are caused by humans and can be dangerous for wildlife and people. To slow their spread, fire retardants are sprayed from great heights onto forests. Through a chemical reaction, these retardants coat trees and other plants with a nonflammable substance called char. This removes access to the fuel—one of three things a fire needs to burn.

FIRE TRIANGLE

A fire needs three elements to burn: fuel, heat, and oxygen—as shown by the sides of this triangle. If any one of these is taken away, the fire will not be able to burn.



How fire retardants work

For a forest fire to be tackled successfully, the retardant must be sprayed onto vegetation before it catches fire. The substances in the retardant react with a chemical called cellulose in the plants. When fire reaches the plants, the heat breaks down the products of this chemical reaction, forming a protective coating of char—a form of carbon. This stops the fire from getting to the fuel it needs to burn, preventing it from spreading further.

